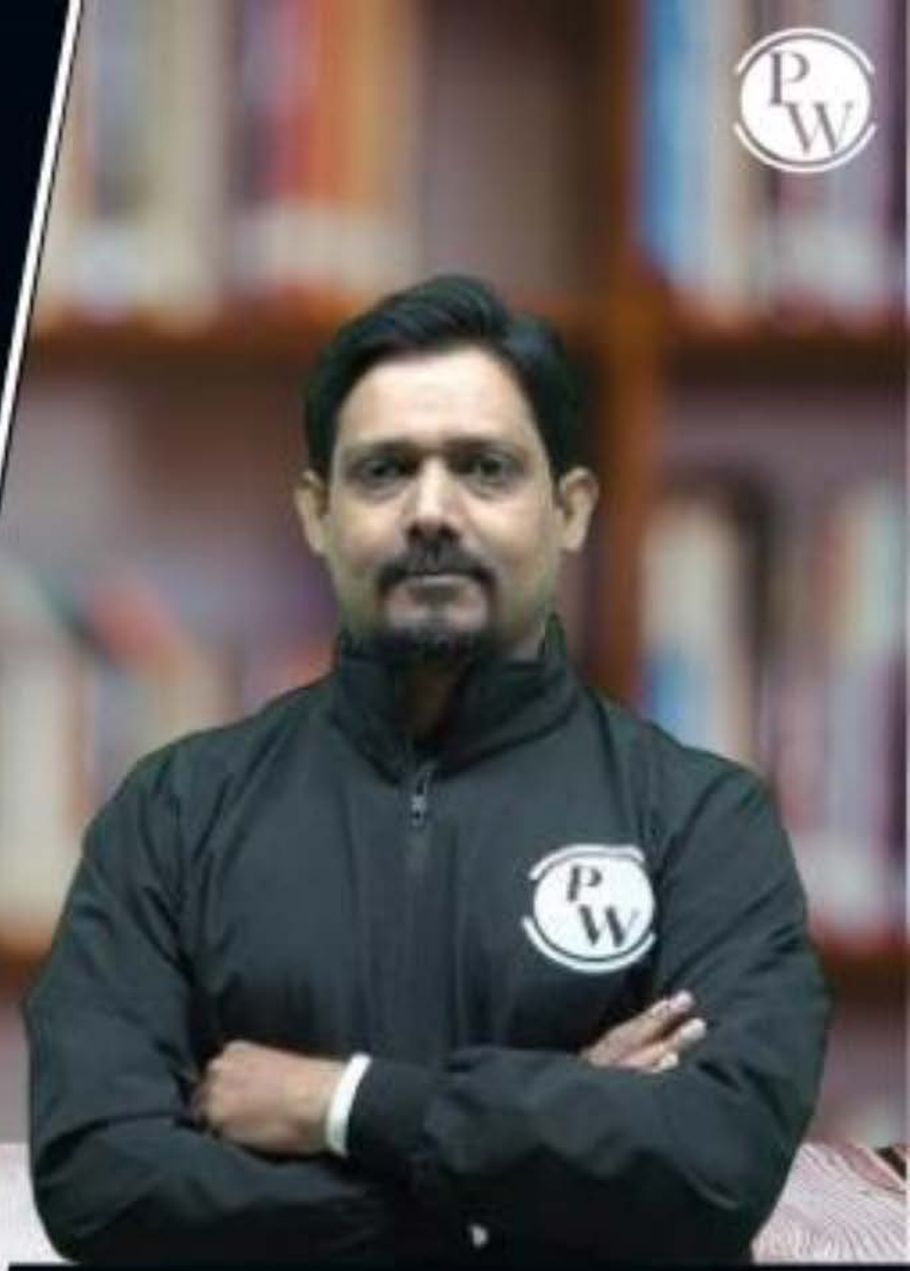
Computer Science & IT



Data Structure & Programming

Tree

Lecture No. 04



By- Abhishek Sir

Recap of Previous Lecture



Topic

Topic

Topic

Topic

Topic

Theorem $n_0 = n_2 + 1$

Binary Tree Representation

Traversal

Slide

Topics to be Covered



- Topic — Traversal
- Topic — Construction of binary tree traversal
- Topic — BST ✓
- Topic
- Topic

Slide

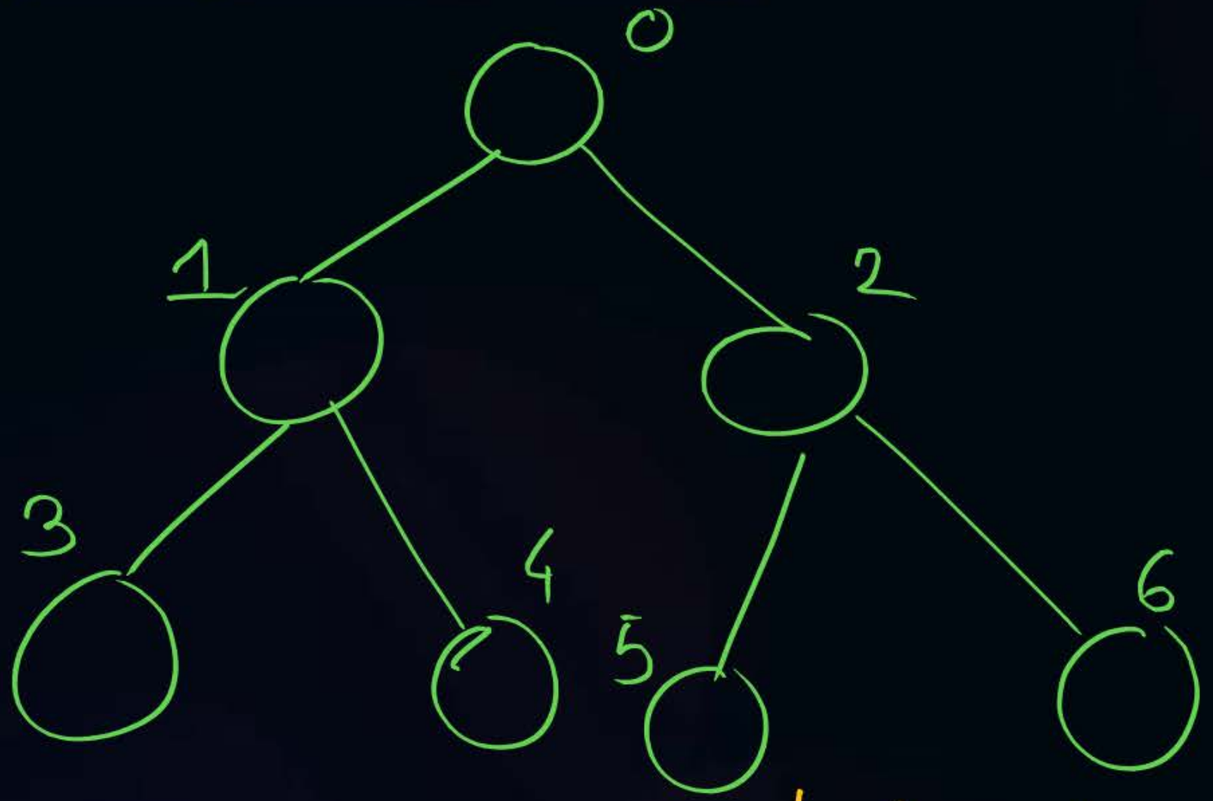


Topic : Tree

Traversal
Construction of Binary tree from traversal
Tree Recursive problem



Topic : Tree



$\left\lfloor \frac{(i-1}{2} \right\rfloor$

i is Index of a Node

$(i \neq 0)$

Leftchild : $2i+1$

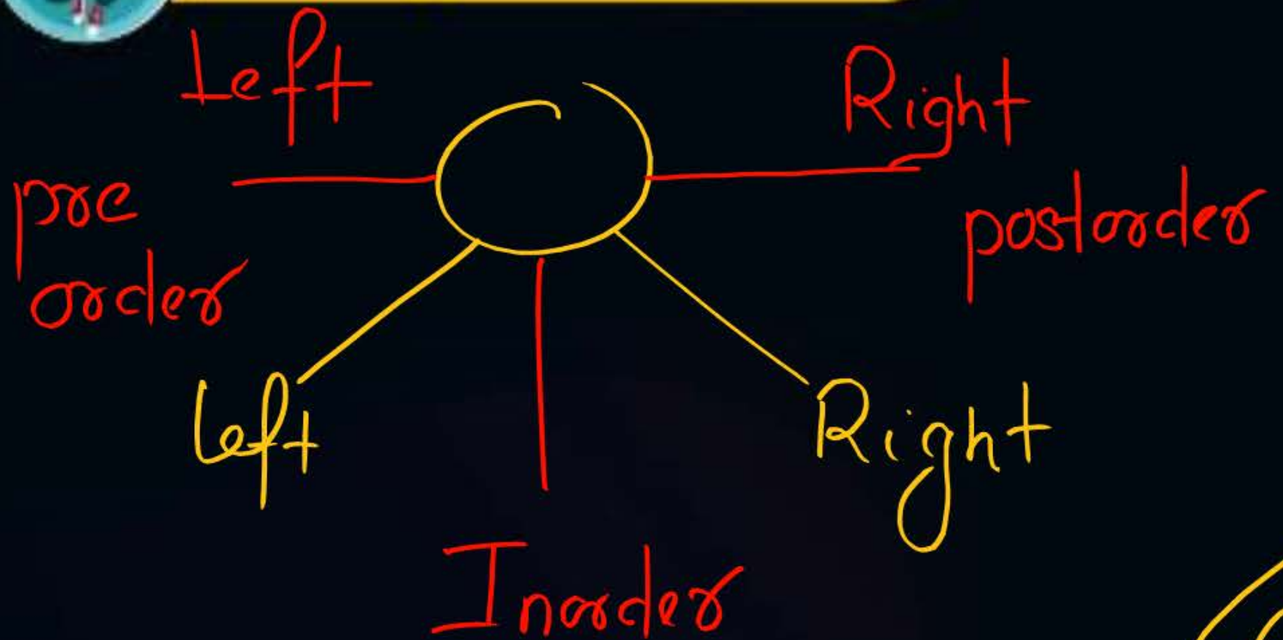
Rightchild : $2i+2$

parent index : $\left\lfloor \frac{((i-1))}{2} \right\rfloor$

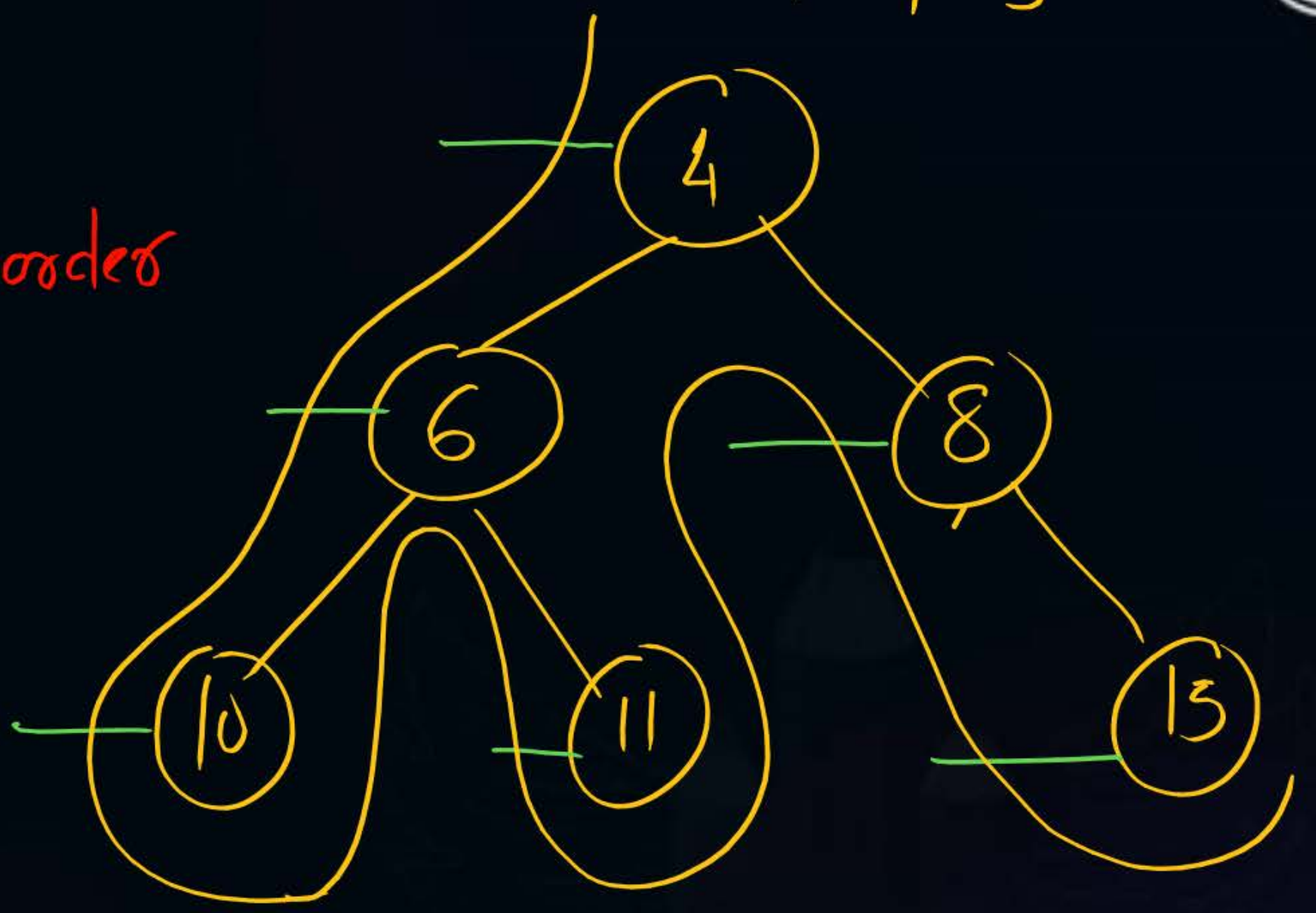
Slide



Topic : Tree



4, 6, 10, 11, 8, 15

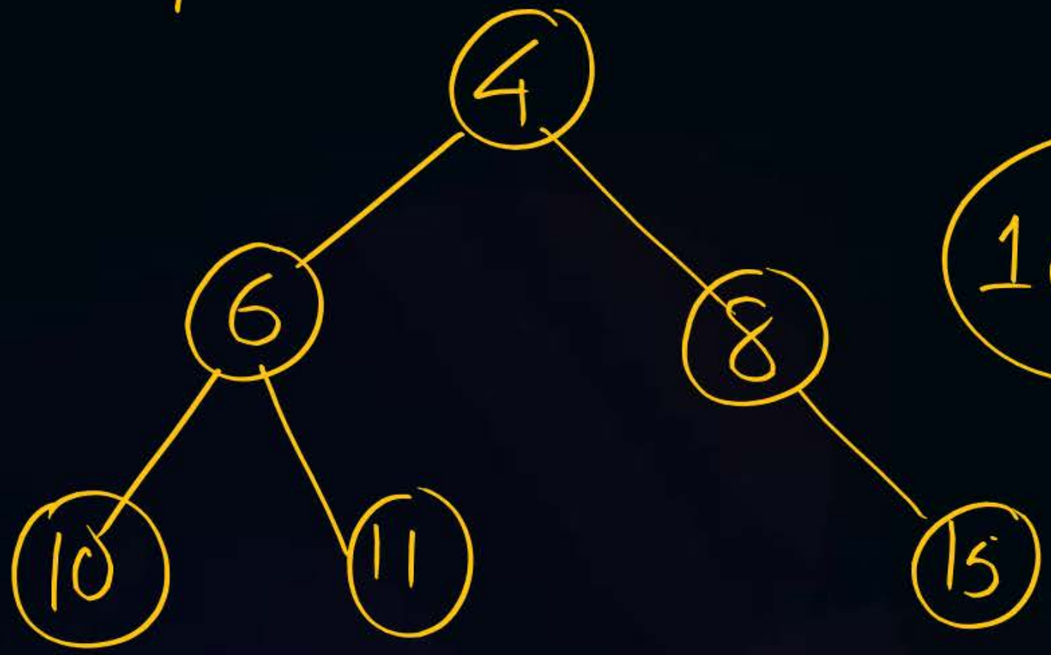


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Topic : Tree

postorder

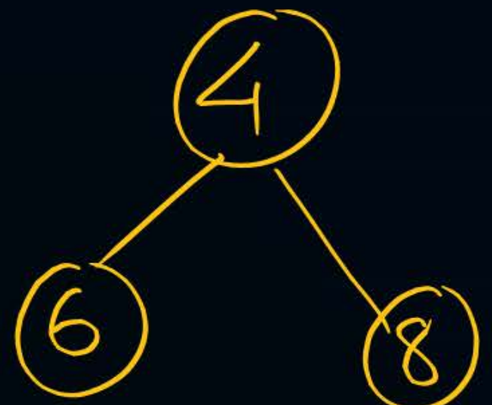


Left Right Root

Left

Right

10 11 6 15 8 4



6 8 4

Slide



In postorder traversal Root is Last one to be visited.

In preorder traversal Root is first Node to be visited.



Topic : Tree

10, 11, 6, 15, 8, 4



Slide



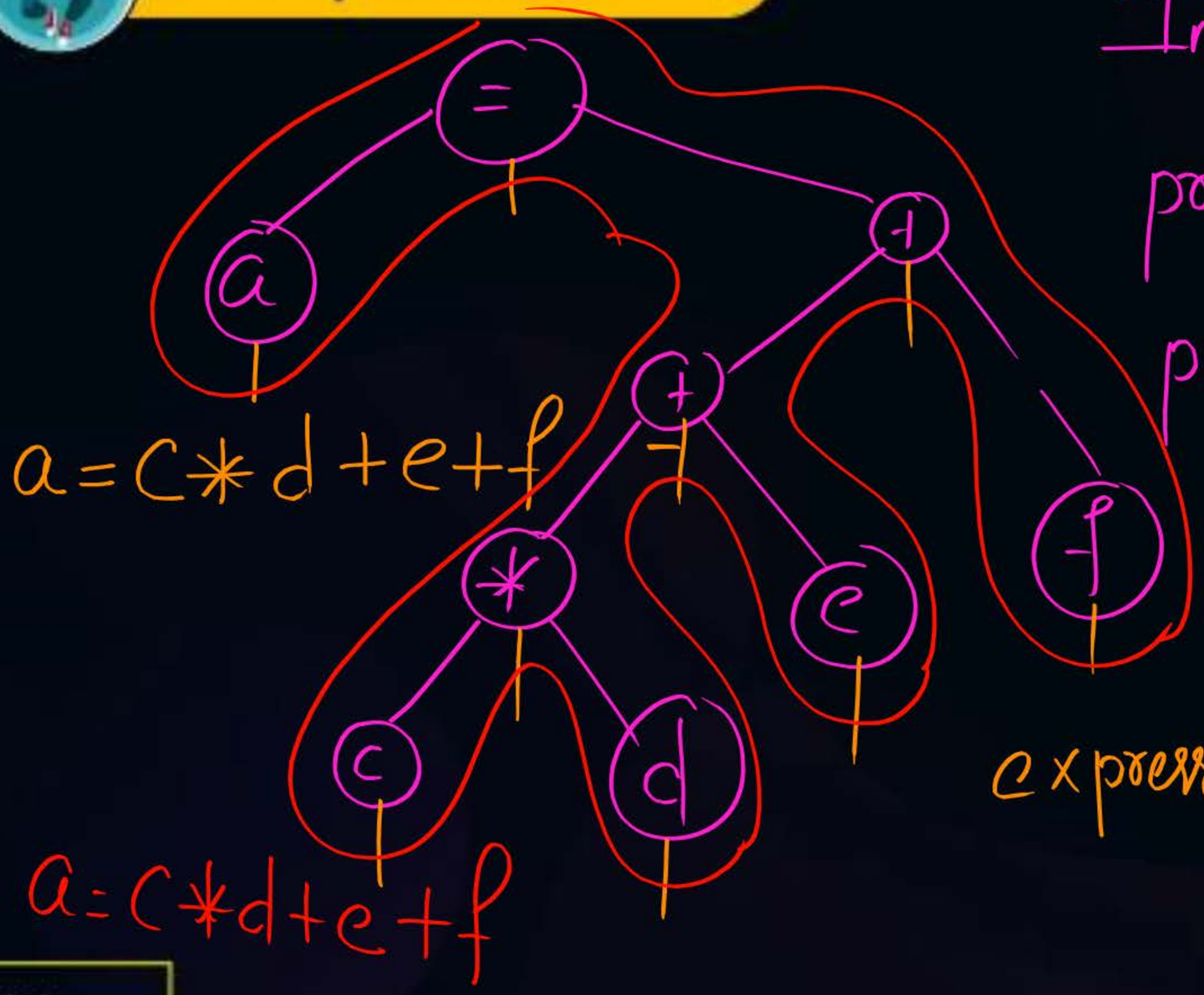
Topic : Tree

Lowest precedence (∈)

Inorder: a = c*d+e+f

preorder: =a++*cdef

postorder: acd*e+f+=

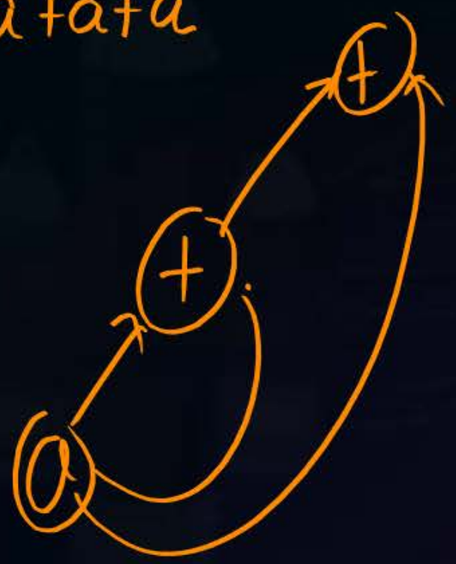


a=c*d+e+f

a=c*d+e+f

expression

a+a+a



DAG

Slide



Topic : Tree

Construction of Binary tree with Inorder & preorder

preorder: a b c d f g e

Inorder: c b f d g | a | e

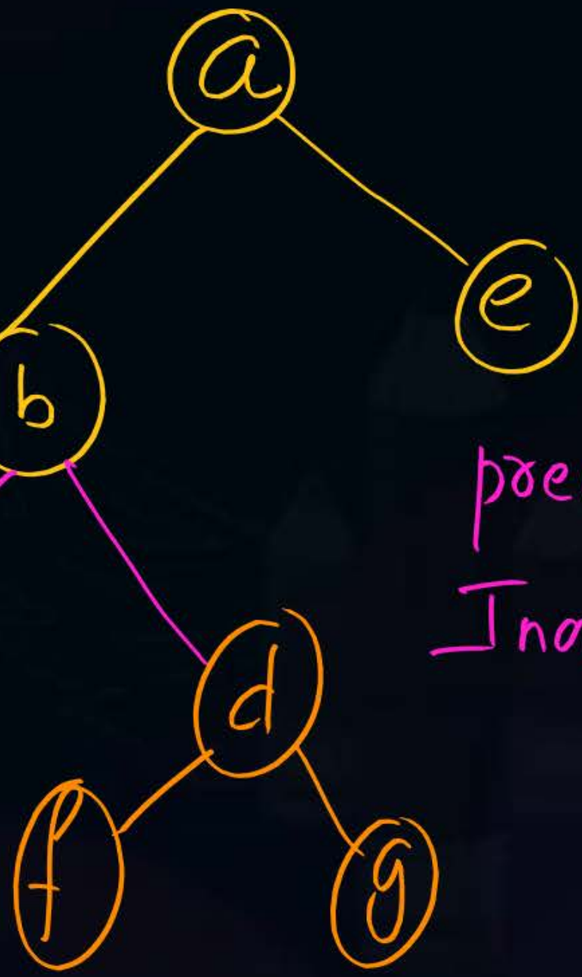
Left | Root | Right

preorder: b c | d f g

Inorder: d b f | d g

preorder: d f g

Inorder: f | d | g



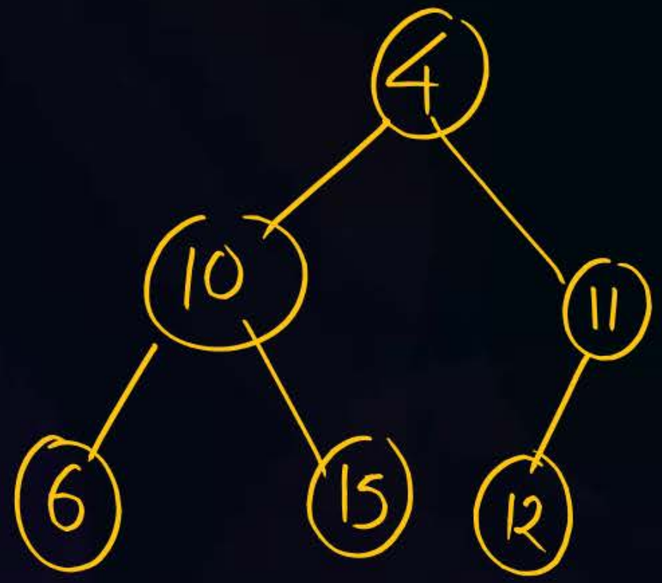
Slide



Topic : Tree

Inorder: 6, 10, 15 | 4 | 12, 11

preorder: 4, 10, 6, 15 | 11, 12





In: 6, 10, 15

pre: 10, 6, 15

In: 12, 11

pre 11, 12



Slide



Topic : Tree

1. find Root in preorder
2. Take the Root in Inorder traversal & split in left and right Subtree.
3. if single Node exists then draw it and repeat the process

Slide



Topic : Question

#Q.

For given preorder : ABCDEFG

inorder : BDCAFGE

What is the postorder of a *Binary Tree*?

(a) CDBGFEA (b) DBCGFEA

(c) DCBGFEA (d) None of these

A B C D E F G - pre

B D C A F G E - Inorder

A

B C D pre

B D C : In

E F G : pre

F G E : In

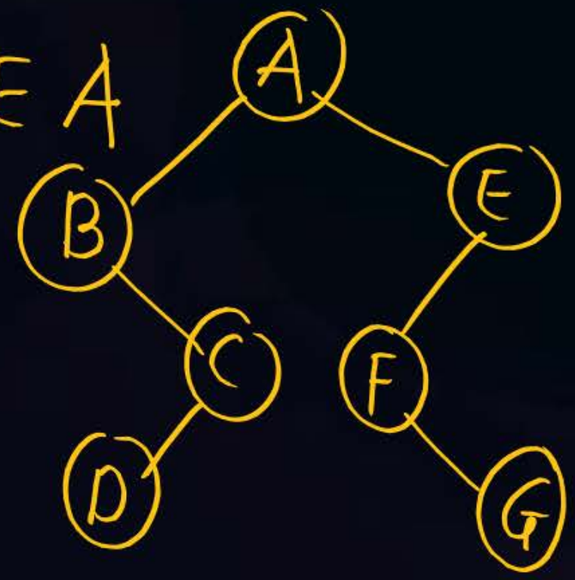
pre : C D

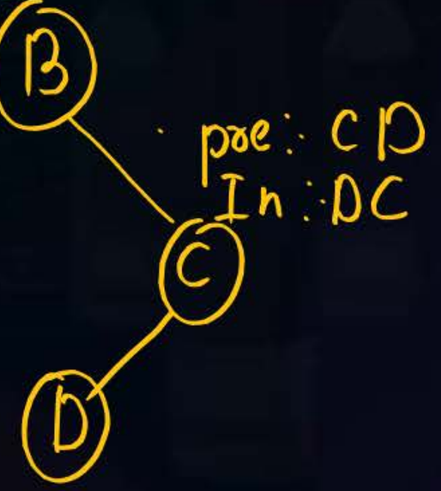
In : D C

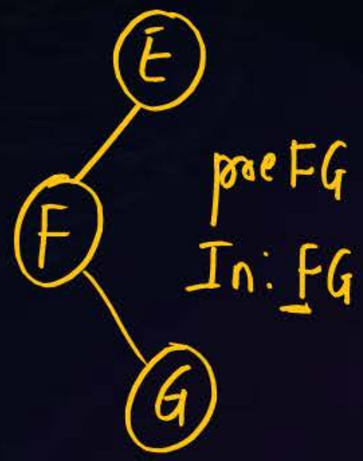
pre FG

In : FG

D C B G F E A







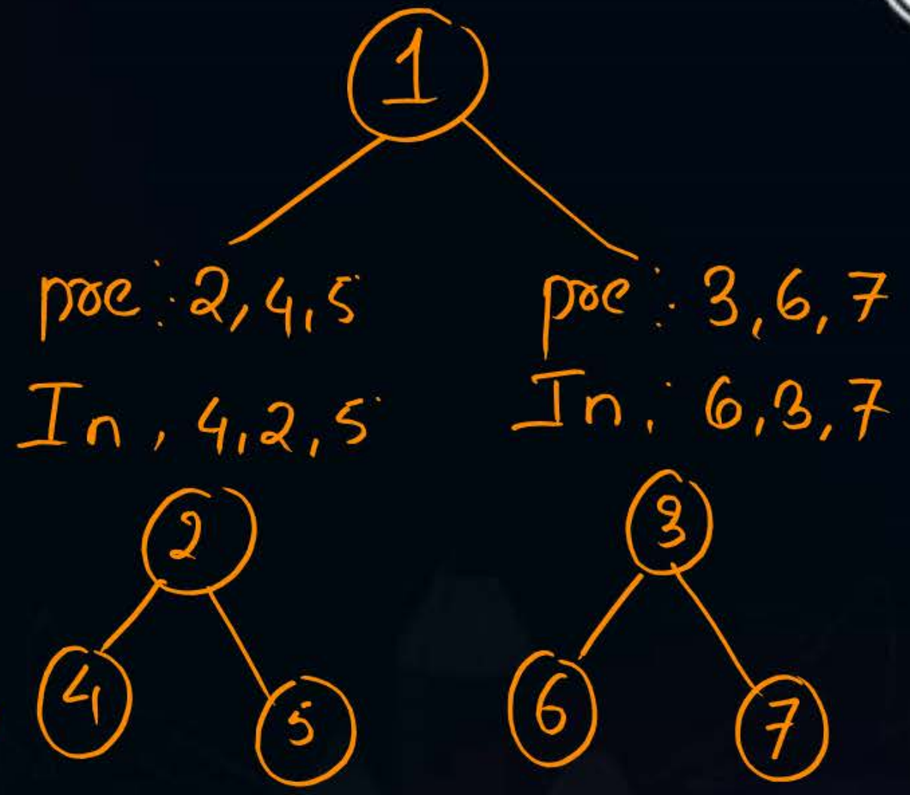
Slide



Topic : Tree

pre 1 | 2, 4, 5 | 3, 6, 7

In : 4, 2, 5 | 1 | 6, 3, 7



postorder traversal

(A) 4, 5, 1, 2, 6, 7, 3 ✗

(B) 4, 5, 2, 1 6, 3, 7 ✗

(C) 4, 5, 2 6, 7, 3, 1

(D) 4, 5, 2, 7, 6, 3, 1

4, 5, 2, 6 7, 3 1

Slide



Topic : Tree

postorder: c f g d b e a

Inorder : c b f, d, g, a, e

$10^{x} + 2^{y} + 3^{z}$

is _____

if x is height

y is No. of Node in Left subtree

z is No of Node in Right subtree

Slide



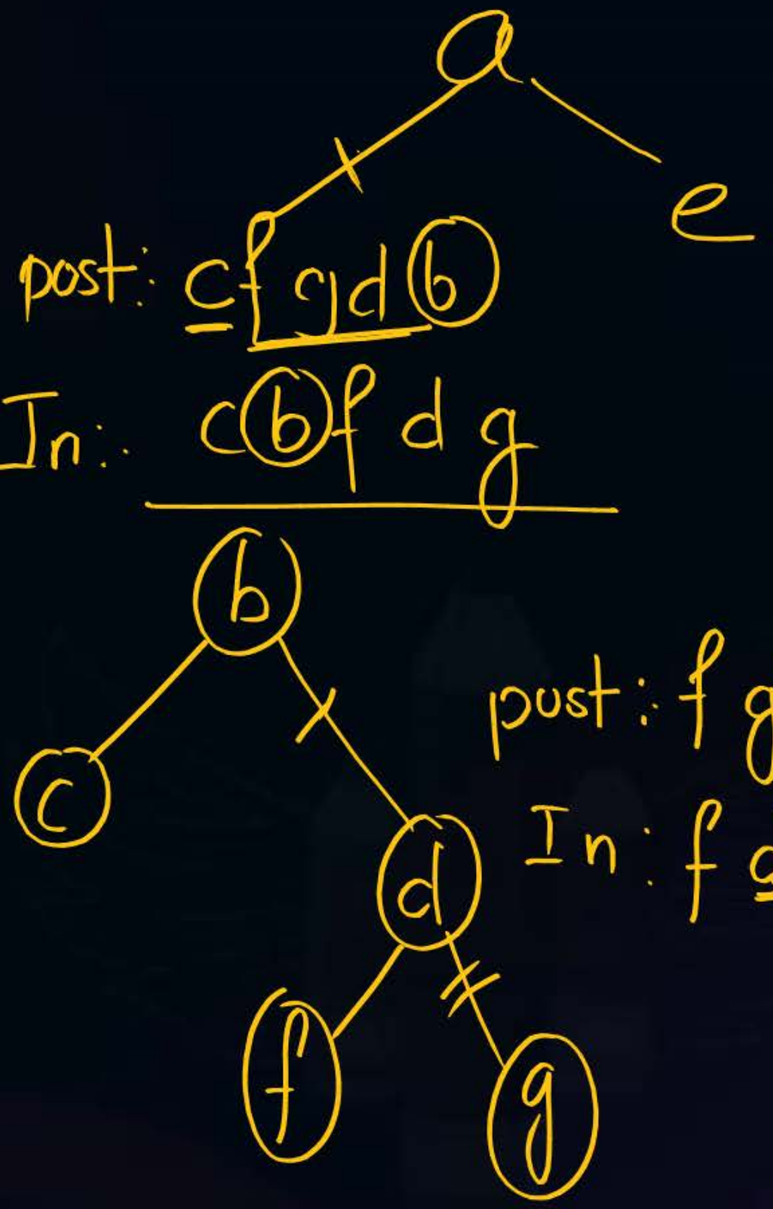
Topic : Tree

postorder: c f g d b e @

Inorder : c b f d g @ e

$10 \times 3 + 2^5 + 3 \times 1$

$30 + 32 + 3 = 65$



Slide



Topic : Tree

Special case

preorder, postorder Inorder

Slide



Topic : Tree

Binary Search Tree (BST)

In binary search tree all values in left subtree is lesser than root and all values in right subtree greater than root. This is true Recursively for subtree also.

Slide



Topic : Tree

Insert the following keys in Empty BST

9, 4, 2, 10, 15, 8, 7, 21, 14



Slide



Topic : Tree

Construct binary Search by inserting following elements

10, 16, 4, 12, 3, 6, 5, 1, 9

if No. of leaves is $\overset{4}{x}$ and height is $\overset{3}{y}$ then x^y is ?

(10)

$4^3 = (64)$

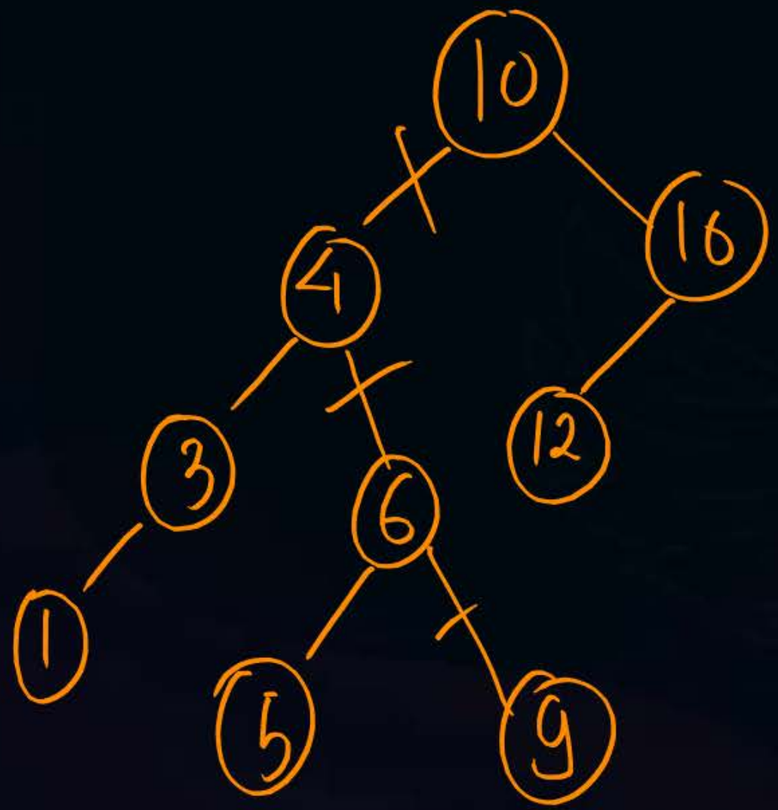
Slide



Topic : Tree

Construct binary Search by inserting following elements

10, 16, 4, 12, 3, 6, 5, 1, 9



Height 3

Leaf Node : 4

Slide



Topic : Tree

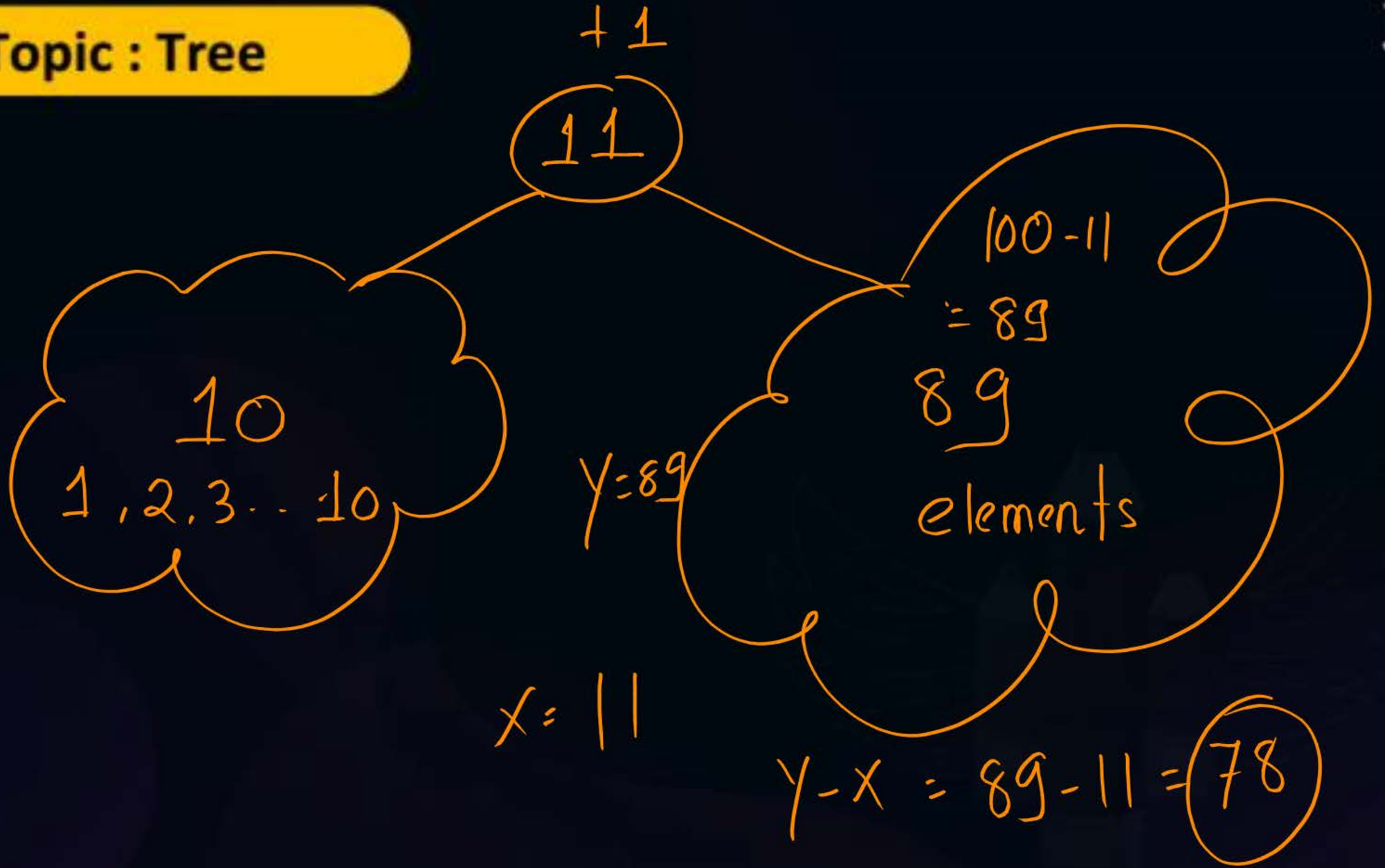
Basic #

if Number 1-100 are Inserted in empty BST and there are 10 element in left Subtree then if x is Root value and y is No. of Node in Right subtree the y - x is _____

Slide



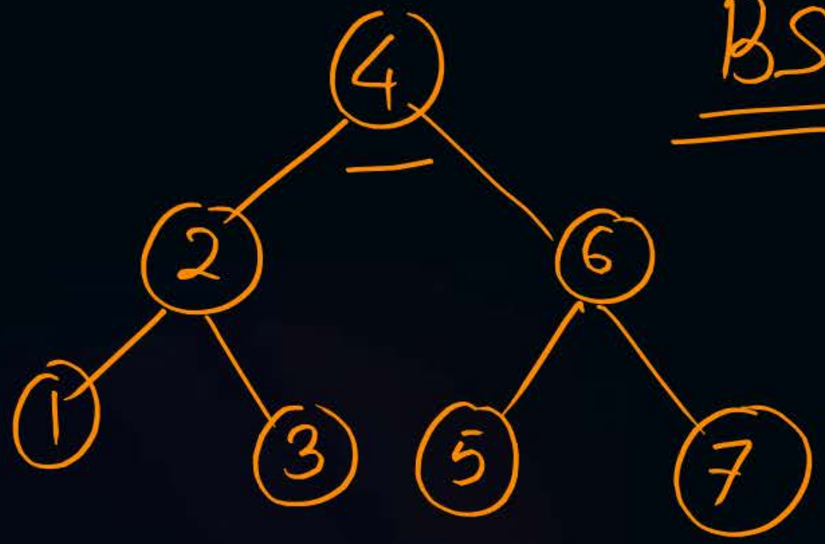
Topic : Tree



Slide



Topic : Tree



BST Inorder traversal

Left Root Right

Lesser Greater

1 2 3 4 5, 6, 7 ←

BST Inorder is Increasing order Sequence (unique element)

Slide



2 mins Summary

Topic — Traversal

Topic — Construction of binary tree (preorder - Inorder)

Topic — (postorder - Inorder)

Topic — BST start

Topic

Slide

THANK - YOU